

Impact of Non-IID Heterogeneity on Federated IoT Intrusion Detection: A CICIOT2023 Study

Satyam Dwivedi¹, Oluseyi Olukola¹, and Nick Rahimi¹

School of Computing Sciences and Computer Engineering,
The University of Southern Mississippi,
Hattiesburg, MS, USA
{satyam.dwivedi,oluseyi.olukola,nick.rahimi}@usm.edu

Abstract. Federated learning (FL) has been proposed as a privacy-preserving approach to collaborative intrusion detection in distributed IoT environments, yet its reliability under non-IID client data distributions remains insufficiently understood. This paper presents a controlled measurement study on the CICIOT2023 dataset, evaluating FedAvg under IID, moderate non-IID ($\alpha = 0.3$), and strong non-IID ($\alpha = 0.1$) Dirichlet partitions across three random seeds. Our results suggest that macro-F1 may drop from 0.6242 ± 0.0049 under IID to 0.3417 ± 0.0059 under strong heterogeneity, with apparent collapse in BruteForce and Benign class detection despite nontrivial aggregate accuracy. A 1D CNN sensitivity experiment and a FedProx baseline further suggest that this degradation pattern may not be architecture-specific and may not be resolved by proximal regularization alone. These observations indicate that accuracy-only evaluation could be insufficient for assessing federated IoT intrusion detection reliability, and that per-class, multi-seed reporting may be necessary to surface class-level failures in practice.

Keywords: Federated learning · Intrusion detection · IoT security · Non-IID data · CICIOT2023 · Class imbalance · FedAvg · FedProx

1 Introduction

Internet of Things (IoT) networks are increasingly deployed in environments where traffic data is distributed across heterogeneous devices, gateways, and administrative domains. Intrusion detection systems (IDSs) operating in such environments must contend with diverse traffic patterns while respecting practical constraints such as data privacy, limited bandwidth, and regulatory boundaries. Federated learning (FL) has emerged as a promising approach in this context, as it allows multiple clients to collaboratively train a shared model without directly pooling raw traffic data [13].

Despite this promise, FL-based intrusion detection in realistic IoT deployments may face a fundamental challenge: statistical heterogeneity. When different clients observe substantially different distributions of benign traffic and attack types, locally computed model updates can diverge in ways that degrade

global model quality [10]. It remains unclear, however, to what extent this heterogeneity affects class-balanced detection reliability rather than aggregate accuracy alone. A model may retain acceptable overall accuracy while silently failing to detect minority but security-critical attack classes – a distinction that aggregate metrics tend to obscure.

A related gap concerns evaluation rigour. Much of the existing FL-IDS literature appears to rely on aggregate accuracy as the primary performance measure, and relatively few studies report per-class F1 behaviour across multiple random seeds under varied non-IID conditions. Without such reporting, it is difficult to assess whether a federated IDS would remain reliable in practice, particularly for rare attack categories that may only appear in a small subset of clients under strong label skew.

This paper attempts to address these gaps through a controlled measurement study on CICIOT2023 [16], a large-scale IoT intrusion detection benchmark, using an 8-category attack formulation (cat8). Rather than proposing a new aggregation method or poisoning defense, our aim is to provide a careful baseline characterisation of how benign non-IID label-distribution skew affects class-balanced FL-IDS reliability under a rigorous evaluation protocol. We first establish centralized baselines using both classical machine learning and deep learning models to anchor subsequent federated comparisons. We then evaluate a FedAvg-based multilayer perceptron (MLP) under three client partition conditions – IID, moderate non-IID (Dirichlet $\alpha = 0.3$), and strong non-IID (Dirichlet $\alpha = 0.1$) – across three random seeds. To explore whether any observed degradation may be architecture-specific, we further include a FedAvg-1D CNN sensitivity experiment, and we report FedProx results as a standard heterogeneity-aware optimizer baseline.

To the best of our knowledge, existing CICIOT2023 FL-IDS studies have not jointly provided: (i) a multi-seed evaluation protocol, (ii) an imbalanced cat8 test distribution, (iii) per-class F1 collapse analysis across IID and non-IID partitions, and (iv) an architecture-sensitivity check within a single controlled study. The combination of these elements, rather than any single component alone, may be what makes it possible to observe the class-level reliability failures that this study documents.

The contributions of this paper are as follows:

- We establish centralized CICIOT2023 cat8 baselines under an imbalanced holdout evaluation protocol, suggesting that aggregate accuracy alone may be insufficient for assessing minority-class detection capability.
- We evaluate FedAvg under IID and Dirichlet-based non-IID partitioning across three random seeds, covering moderate ($\alpha = 0.3$) and strong ($\alpha = 0.1$) label-distribution skew, and find evidence of substantial macro-F1 degradation with increasing heterogeneity.
- We conduct a FedAvg-1D CNN architecture-sensitivity experiment, providing preliminary evidence that the observed class-collapse pattern may not be specific to the MLP architecture.

- We find that per-class F1 can collapse for security-relevant classes – including BruteForce, Spoofing, and Benign – under strong non-IID conditions, even when aggregate accuracy remains nontrivial.
- We include FedProx ($\mu = 0.01$) as a supporting baseline and observe that proximal regularization alone appears insufficient to resolve class-level collapse under strong label skew.
- We propose a multi-seed, per-class, imbalanced-test evaluation protocol for FL-IDS studies, arguing that such a protocol may be necessary to surface reliability failures hidden by accuracy-based reporting.

The remainder of this paper is organized as follows. Section 2 reviews related work on centralized and federated IoT intrusion detection and the challenges of non-IID heterogeneity. Section 3 describes the dataset, experimental protocol, and federated learning setup. Section 4 presents the centralized baseline results. Section 5 reports the federated learning results including architecture sensitivity and FedProx comparisons. Section 6 discusses the main findings and practical implications. Section 7 outlines the study’s limitations, and Section 8 concludes the paper.

2 Related Work

2.1 Centralized IoT Intrusion Detection

Centralized intrusion detection on IoT traffic datasets has been widely studied using classical machine learning and deep learning methods. Modern IoT and network intrusion datasets, including CICIoT2023 [16], Edge-IIoTset [7], and NetFlow-based benchmarks [23], support binary and multiclass evaluation across diverse attack families. Prior work often reports strong aggregate performance from tree-based ensembles, gradient boosting, and neural models. Ensemble strategies have shown particular promise for IDS accuracy, combining multiple base classifiers to improve detection rates [1]. However, aggregate metrics can hide poor behavior on minority or low-support attack classes, including zero-day attack categories that exhibit limited training support [6]. For this reason, class-wise metrics and macro-F1 are important when evaluating multiclass IDS models under imbalanced traffic distributions. Vulnerabilities in web-based applications further underscore the need for robust, multi-layered detection [20].

2.2 Federated Learning for IoT Intrusion Detection

Federated learning has been proposed as a privacy-preserving and distributed alternative to centralized training for IoT intrusion detection [13]. Existing FL-IDS studies generally show that FL can achieve competitive performance while avoiding direct raw-data sharing. Recent work has also begun to evaluate FL-IDS under realistic non-IID client distributions, client-level traffic skew, and poisoning conditions. For example, FedMADE studies robust federated intrusion detection on CICIoT2023 using dynamic clustering based on class probability matrices [24],

while pFL-IDS and AGAT-FL explore personalized or trust-aware defenses for heterogeneous and adversarial IoT settings [22,25]. These studies show that the field has moved beyond generic FedAvg evaluation toward robust and adaptive FL-IDS mechanisms. Related adversarial robustness work from the same lab has shown that adversarially trained classifiers can substantially improve accuracy under crafted input attacks [9], and that multi-stage, attack-aware defense frameworks provide meaningful robustness gains for network intrusion detection under gradient-based and distribution-shift threat models [17]. Sequential LSTM-based detection approaches have also demonstrated strong performance for insider threat scenarios characterized by temporal behavioral patterns [18].

2.3 Non-IID Heterogeneity and Robustness

Non-IID data is a known challenge in federated learning [10]. Methods such as FedProx and SCAFFOLD address client drift and optimization instability under heterogeneous data [11,12]. In security settings, robust aggregation methods such as Krum, coordinate-wise median, trimmed mean, and trust-based aggregation have been proposed to mitigate Byzantine or poisoning behavior [3,4]. However, many such methods are evaluated on image, text, or general FL benchmarks, while IoT intrusion detection introduces additional class imbalance and security-critical minority-class concerns. IoT environments in particular are characterized by heterogeneous device behaviors and attack surfaces, including phishing [14] and side-channel threats [19], motivating the need for federated detection approaches that remain reliable under realistic data heterogeneity. Distributed P2P fog architectures offer one infrastructure model for such deployments [15].

2.4 Positioning of This Study

In contrast to work proposing new aggregation methods or poisoning defenses, this paper provides a controlled measurement of how increasing Dirichlet label-distribution skew affects class-balanced reliability in standard FL baselines. Our emphasis on multi-seed macro-F1, weighted-F1, and per-class F1 under an imbalanced cat8 test distribution reveals class-level collapse that is hidden by aggregate accuracy alone.

Table 1 summarizes how this work relates to the most relevant prior studies. Key dimensions include the dataset used, the FL algorithm evaluated, whether non-IID heterogeneity is explicitly studied, whether class-imbalance is addressed in evaluation, and whether per-class F1 collapse is reported.

3 Dataset and Experimental Setup

3.1 Dataset

We use CICIoT2023, a large-scale IoT intrusion detection dataset containing benign traffic and multiple attack categories [16]. For the main multiclass experiments, we use the cat8 formulation with the following classes: DDoS, DoS,

Table 1. Comparison of related FL-IDS studies. ✓ = yes; – = no; ~ = partial.

Study	Dataset	Non-IID	Imbal. Eval	Per-Class F1	Multi-Seed
FedMADE [24]	CICIoT2023	~	✓	–	–
pFL-IDS [25]	Mixed IoT	~	–	–	–
AGAT-FL [22]	Mixed IoT	–	–	–	–
Dahal et al. [6]	NSL-KDD	–	✓	~	–
Baluguri et al. [1]	MQTT/IoT	–	✓	–	–
This work	CICIoT2023	✓	✓	✓	✓

Mirai, Benign, Recon, Spoofing, Web, and BruteForce. This formulation provides a middle ground between binary attack detection and fine-grained attack-family classification.

3.2 Train/Test Protocol

Before training-time resampling, the training, validation, and test splits are stratified to preserve the original imbalanced cat8 class proportions. Training uses the prepared training split, while validation and test evaluation preserve the imbalanced class distribution.

Table 2. Cat8 class distribution across train, validation, and test splits.

Class	Train	Val	Test	Test (%)
DDoS	8,604,445	1,229,207	2,458,413	58.52
DoS	2,925,190	417,885	835,769	19.89
Mirai	1,651,428	235,918	471,837	11.23
Benign	733,115	104,731	209,462	4.99
Recon	458,825	65,546	131,093	3.12
Spoofing	305,243	43,606	87,212	2.08
Web	16,595	2,371	4,741	0.11
BruteForce	8,764	1,252	2,504	0.06

Table 2 shows that the test set is highly imbalanced: DDoS and DoS account for approximately 78.4% of test samples, while Web and BruteForce together account for less than 0.2%. This motivates the use of macro-F1 and per-class F1 in addition to accuracy.

To reduce extreme training imbalance during model fitting, the training split was balanced using SMOTE-ENN before centralized and federated training [2, 5]. Validation and test splits were not resampled and preserve the original imbalanced cat8 class proportions. All preprocessing artifacts were fit using training data only and applied consistently to validation and test splits.

3.3 Stage 1: Centralized Baselines

For the centralized setting, we evaluate Random Forest (RF), XGBoost, MLP, and 1D CNN models. These baselines provide both classical machine learning and neural references for the cat8 task. The MLP is later used as the primary neural architecture for the federated experiments so that centralized and federated neural results are directly comparable.

3.4 Stage 2: Federated Learning Setup

For federated learning, we use 10 simulated clients, 20 communication rounds, 3 local epochs, and batch size 4096. We evaluate three client partition settings: IID, Dirichlet non-IID with $\alpha = 0.3$, and Dirichlet non-IID with $\alpha = 0.1$, following common Dirichlet label-skew evaluation practice [8]. Unless otherwise stated, FL results are reported as mean $\pm$ standard deviation across three seeds: 123, 2026, and 3407.

The primary federated architecture is an MLP consisting of an input layer matching the feature dimension, followed by dense layers with 256, 128, and 64 units using ReLU activations. Batch normalization is applied after the 256-unit and 128-unit layers. Dropout with rate $p = 0.30$ is applied after the first two hidden layers, and dropout with rate 0.20 after the third. The final layer uses softmax activation over the eight cat8 classes. Both FedAvg and FedProx use Adam with learning rate 10^{-4} and sparse categorical cross-entropy.

To test whether the observed class-collapse pattern is specific to the MLP architecture, we also evaluate FedAvg using a 1D CNN under the same client partitions, communication rounds, local epochs, batch size, and random seeds.

Figure 1 illustrates the client-class distribution under the strongest non-IID setting ($\alpha = 0.1$), showing severe class skew across clients.

3.5 Stage 2B: FedProx Baseline

In addition to FedAvg, we implement FedProx as a supporting heterogeneity-aware optimizer baseline [12]. FedProx adds a proximal penalty to the local client objective, discouraging local weights from drifting too far from the current global model. The local FedProx objective is:

$$\mathcal{L}_{\text{FedProx}}(w) = \mathcal{L}_{\text{CE}}(w) + \frac{\mu}{2} \|w - w_t\|^2, \quad (1)$$

where $\mathcal{L}_{\text{CE}}$ is the sparse categorical cross-entropy loss, w_t denotes the global model weights at round t , and μ controls the proximal penalty strength. We use $\mu = 0.01$ as an initial baseline value.

4 Centralized Baseline Results

4.1 Overall Performance

Table 3 summarizes the centralized cat8 baseline results. Among the baselines, XGBoost achieves the strongest overall performance. The classical machine learning models outperform the neural models on this tabular IDS task. However, the

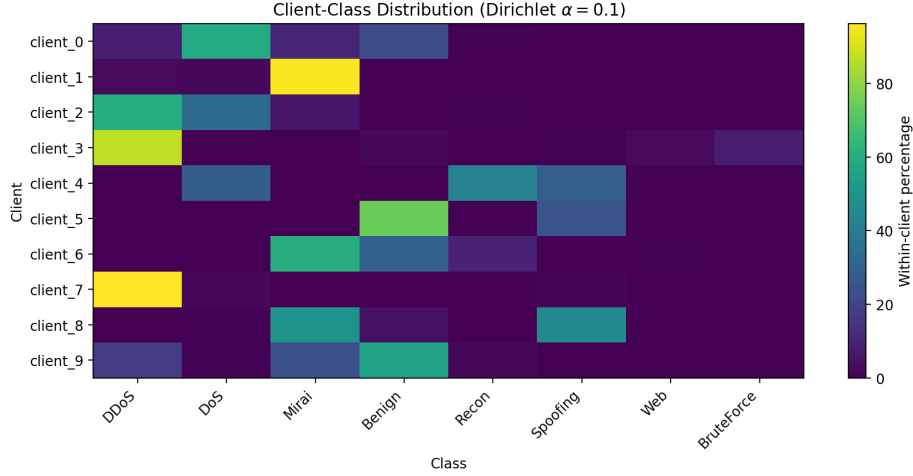


Fig. 1. Client-class distribution heatmap for the strongest non-IID setting ($\alpha = 0.1$). The heatmap shows within-client class percentages and illustrates severe class skew.

MLP remains important as the centralized neural reference for the FedAvg and FedProx experiments.

Table 3. Stage 1 centralized baseline results on the imbalanced cat8 test set.

Model	Accuracy	F1 Macro	F1 Weighted
Random Forest	0.7966	0.6687	0.8079
XGBoost	0.7989	0.6746	0.8101
MLP	0.7439	0.5861	0.7604
1D CNN	0.7316	0.5701	0.7480

4.2 Class-Level Observations

A key observation from Stage 1 is that minority classes remain difficult even in the centralized setting. Under XGBoost, Web achieves F1=0.1770 and BruteForce achieves F1=0.3041, compared with Mirai at F1=0.9960 and Spoofing at F1=0.8878. This shows that minority-class difficulty exists before federated partitioning is introduced.

5 Federated Learning Results

5.1 Overall Multi-Seed FedAvg Comparison

Table 4 reports FedAvg results across three seeds. Under IID partitioning, FedAvg achieves strong performance and exceeds the centralized MLP baseline on all three aggregate metrics. However, macro-F1 drops from 0.6242 ± 0.0049 under IID to 0.5254 ± 0.0083 under $\alpha = 0.3$ and 0.3417 ± 0.0059 under $\alpha = 0.1$, representing a substantially larger relative degradation in macro-F1 than in accuracy.

Table 4. FedAvg multi-seed results under IID and non-IID client partitioning.

Setting	Accuracy		Macro-F1		Weighted-F1	
	Mean	Std	Mean	Std	Mean	Std
IID	0.8282	0.0001	0.6242	0.0049	0.8122	0.0006
$\alpha = 0.3$	0.7983	0.0031	0.5254	0.0083	0.7698	0.0152
$\alpha = 0.1$	0.7673	0.0030	0.3417	0.0059	0.7132	0.0086

5.2 Architecture Sensitivity: FedAvg with 1D CNN

Table 5 shows FedAvg-1D CNN results. The same overall degradation trend appears: macro-F1 decreases from 0.5585 ± 0.0011 under IID to 0.4464 ± 0.0135 under $\alpha = 0.3$ and 0.3601 ± 0.0230 under $\alpha = 0.1$. This suggests that class-balanced reliability degradation is not solely an artifact of the MLP architecture.

Table 5. FedAvg-1D CNN architecture-sensitivity results under IID and non-IID partitioning.

Setting	Accuracy		Macro-F1		Weighted-F1	
	Mean	Std	Mean	Std	Mean	Std
IID	0.8219	0.0003	0.5585	0.0011	0.8023	0.0011
$\alpha = 0.3$	0.7858	0.0157	0.4464	0.0135	0.7607	0.0154
$\alpha = 0.1$	0.7766	0.0052	0.3601	0.0230	0.7338	0.0100

5.3 Key Per-Class Effects under FedAvg

Table 6 highlights four security-critical classes. Web remains weak across all FL settings. BruteForce, Spoofing, and Benign show a direct heterogeneity effect: under $\alpha = 0.1$, BruteForce and Benign fall to zero F1, while Spoofing is nearly

eliminated. A likely mechanism is the interaction between rare-class support and client-level label skew: under strong Dirichlet partitioning, rare classes may appear in very few clients, providing little or no gradient signal on most clients while sample-size-weighted aggregation is dominated by majority-class clients.

Table 6. Key per-class F1 results under FedAvg.

Setting	Web		BruteForce		Spoofing		Benign	
	Mean	Std	Mean	Std	Mean	Std	Mean	Std
IID	0.0421	0.0364	0.3390	0.0029	0.6846	0.0023	0.7973	0.0012
$\alpha = 0.3$	0.0901	0.0034	0.2396	0.0240	0.2376	0.0151	0.7154	0.0012
$\alpha = 0.1$	0.0289	0.0342	0.0000	0.0000	0.0003	0.0005	0.0000	0.0000

5.4 Convergence Behavior

Figure 2 shows validation macro-F1 convergence across communication rounds. IID training reaches the strongest class-balanced performance, $\alpha = 0.3$ converges to a lower macro-F1 regime, and $\alpha = 0.1$ remains substantially weaker throughout training, confirming that the performance gap is not a final-round artifact.

5.5 Supporting FedProx Comparison

Table 7 reports FedProx as a supporting heterogeneity-aware baseline. FedProx performs slightly below FedAvg under IID and $\alpha = 0.3$ partitioning, while the two completed $\alpha = 0.1$ seeds show higher macro-F1 than FedAvg. However, class-level results in Table 8 show that FedProx does not eliminate class collapse – Web remains undetected across all completed FedProx runs.

Table 7. FedProx supporting-baseline results. IID and $\alpha = 0.3$ use three seeds; $\alpha = 0.1$ uses two completed seeds.

Setting	Seeds	Accuracy		Macro-F1		Weighted-F1	
		Mean	Std	Mean	Std	Mean	Std
IID	3	0.8232	0.0008	0.6100	0.0007	0.8058	0.0015
$\alpha = 0.3$	3	0.7996	0.0023	0.5021	0.0028	0.7621	0.0053
$\alpha = 0.1$	2	0.7677	0.0070	0.3919	0.0015	0.7355	0.0021

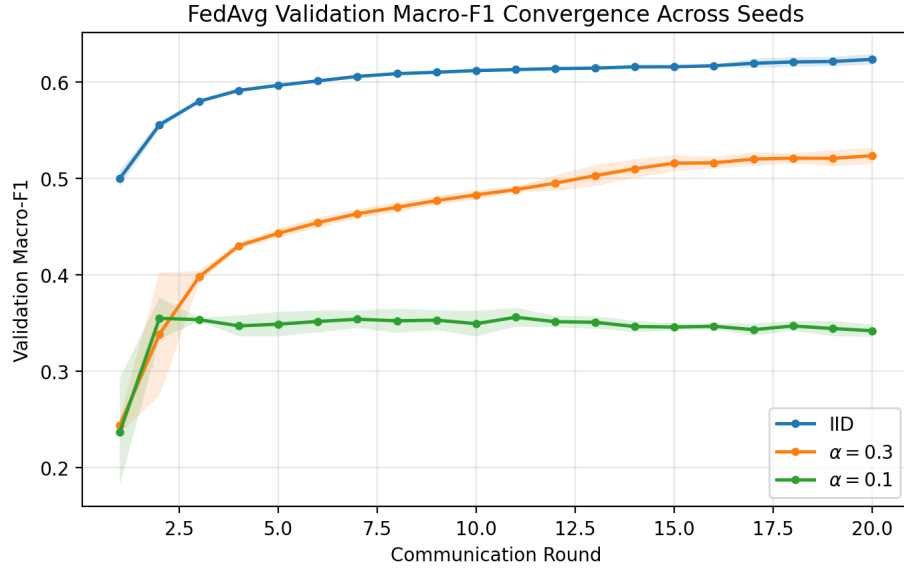


Fig. 2. Validation macro-F1 convergence across FedAvg rounds. Lines show seed-averaged performance and shaded regions show variation across seeds.

Table 8. Supporting key per-class F1 results under FedProx.

Setting	Seeds	Web	BruteForce	Spoofing	Benign
IID	3	0.0000	0.3367	0.6529	0.7892
$\alpha = 0.3$	3	0.0000	0.2320	0.1998	0.6998
$\alpha = 0.1$	2	0.0000	0.1159	0.1727	0.0370

6 Discussion

The results of this study raise several points that warrant interpretation beyond the numerical summaries reported in the preceding sections.

Perhaps the most striking pattern is the divergence between accuracy and macro-F1 as heterogeneity increases. Under strong non-IID partitioning ($\alpha = 0.1$), FedAvg accuracy remains above 0.76, a figure that might appear acceptable in isolation. Yet macro-F1 collapses to 0.3417 ± 0.0059 under the same conditions, and per-class analysis reveals that BruteForce and Benign detection are effectively lost entirely. This gap is not incidental; it appears to reflect how sample-size-weighted aggregation interacts with severe label skew. When rare classes such as BruteForce are concentrated in only a few clients, local training on most clients provides little or no gradient signal for those labels, and the resulting global model may optimise almost entirely for majority-class performance. The aggregate accuracy metric, dominated by DDoS and DoS which together account for approximately 78% of test samples, fails to capture this failure mode. These observations suggest that reporting accuracy alone may give a misleading picture of federated IDS reliability, particularly in the imbalanced traffic distributions that characterise real IoT deployments.

The inclusion of FedProx as a heterogeneity-aware baseline offers a further informative data point. Although FedProx is designed to limit client drift through a proximal penalty term, the results here suggest that this mechanism alone may be insufficient to recover class-balanced performance under strong label skew. Web detection remains near zero across all completed FedProx runs, and Benign detection degrades sharply under $\alpha = 0.1$. This is not necessarily a criticism of FedProx per se; rather, it suggests that the class-collapse problem may require interventions targeted at the class distribution itself, rather than at the optimisation dynamics alone. Approaches such as client-level class-aware sampling, class-weighted local loss, or minority-class replay could plausibly address this more directly, though a systematic comparison of such strategies falls outside the scope of the present study.

The architecture-sensitivity experiment with FedAvg-1D CNN provides some preliminary evidence that the degradation pattern observed with the MLP is not merely an artefact of that particular model’s inductive bias. The 1D CNN exhibits a qualitatively similar macro-F1 decline across IID, $\alpha = 0.3$, and $\alpha = 0.1$ conditions, with Web and BruteForce remaining weak or collapsing entirely under strong heterogeneity. This finding should be interpreted cautiously given that only two architectures were compared; however, it does suggest that the reported class-collapse behaviour may be attributable primarily to the data partitioning conditions rather than to model-specific factors.

Taken together, these results point to a gap in how FL-IDS systems are typically evaluated. Reporting multi-seed macro-F1 and per-class F1 alongside accuracy, and doing so under an imbalanced test distribution that reflects realistic traffic proportions, may reveal reliability issues that would otherwise remain hidden. Related work on adversarial robustness in intrusion detection [9, 17] and sequential threat modelling [18] suggests that the broader field is increasingly

aware of the gap between reported accuracy and operational reliability. Whether the evaluation protocol proposed here could be adopted more broadly in FL-IDS research remains an open question, though the evidence from this study suggests it may be warranted.

7 Limitations and Future Work

Several limitations of this study should be noted alongside directions that future work could reasonably pursue. The federated experiments focus on an MLP as the primary architecture, chosen to match the centralized neural baseline, though this means that the strongest centralized model, XGBoost, is not directly evaluated in the federated setting. Whether a federated gradient-boosted model would exhibit similar or different class-collapse patterns under non-IID conditions remains an open question. The architecture-sensitivity experiment with the 1D CNN provides only partial coverage, and a more systematic architecture comparison would strengthen the generalisability of the findings.

A further limitation concerns the scope of heterogeneity modelled. This study addresses label-distribution skew through Dirichlet partitioning, which is a common and tractable approximation of non-IID conditions, but does not capture other practically relevant forms of heterogeneity. Feature shift arising from different device types, temporal drift as network behaviour evolves, variable client participation rates, and systems heterogeneity such as client dropout are all absent from the current setup. Future work could extend the experimental framework to incorporate these dimensions, likely revealing additional stress points not captured here.

The FedProx evaluation is also constrained in scope. The proximal penalty $\mu = 0.01$ was selected as an initial baseline value and was not tuned; a more complete sensitivity analysis across a range of μ values would be needed before drawing strong conclusions about FedProx’s potential under this dataset and partitioning scheme. Similarly, the $\alpha = 0.1$ FedProx results are based on two completed seeds rather than three due to computational cost, which means the corresponding estimates carry additional uncertainty relative to the IID and $\alpha = 0.3$ results.

Beyond these specific limitations, the study does not compare robust aggregation methods, personalised FL strategies, secure aggregation protocols, or poisoning-defense mechanisms. These represent meaningful extensions that would be necessary for a comprehensive evaluation of federated IDS under adversarial conditions. Cross-dataset validation on benchmarks such as Edge-IIoTset [7] or NetFlow-based datasets [23] would also help establish whether the reported class-collapse patterns generalise beyond CICIOT2023. Finally, the practical mitigation directions suggested in the discussion, including class-aware sampling, class-weighted local loss, and minority-class replay, alongside more principled defense frameworks drawing on related work [17, 21], represent natural candidates for follow-on investigation.

8 Conclusion

This paper presented a controlled empirical study of centralized and federated IoT intrusion detection on CICIOT2023. We established centralized cat8 baselines and evaluated FedAvg under IID and increasingly non-IID client partitioning. Across multiple seeds, FedAvg performs strongly under IID partitioning, but class-balanced performance degrades substantially as heterogeneity increases. Under strong non-IID heterogeneity, several important classes collapse despite nontrivial overall accuracy. The additional FedAvg-1D CNN sensitivity experiment shows a similar degradation pattern, confirming that the observed reliability issue is not limited to the MLP federated architecture. Supporting FedProx results show that adding a proximal penalty does not fully eliminate class-level collapse. Overall, these results indicate that accuracy-only evaluation can mislead operators about federated IDS reliability, and that heterogeneity should be treated as a first-order trustworthy-AI concern in federated IoT IDS evaluation. Future work will extend this study toward poisoning attacks, robust aggregation, personalized FL, and cross-dataset validation, building on adversarial defense and multi-stage detection methods developed in related work [1, 17].

Acknowledgment

This work is part of an ongoing research effort on federated intrusion detection in heterogeneous IoT environments at the University of Southern Mississippi.

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